AI-BASED SOCIAL INTEGRATION AND INCLUSION PLATFORM FOR UNIVERSITY FRESHERS

Project ID: 25-26J-154

Project Proposal Report

Pathiraja H.N. – IT22142214

B.Sc. (Hons) Degree in Information Technology Specializing in Information Systems Engineering

Department of Computer Systems and Network Engineering

Sri Lanka Institute of Information Technology,
Sri Lanka

August 2025

ADAPTIVE STUDENT PROFILING AND LEARNING PERSONA MANAGEMENT

Project ID: 25-26J-154

Project Proposal Report

Pathiraja H.N. – IT22142214

Supervisor: Ms. Narmada Gamage

Co - Supervisor: Ms. Chathurangika Kahandawarachchi

B.Sc. (Hons) Degree in Information Technology Specializing in Information Systems Engineering

Department of Computer Systems and Network Engineering

Sri Lanka Institute of Information Technology,
Sri Lanka

August 2025

Declaration of the Candidate & Supervisor

Declaration of the Candidate & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
H. N. Pathiraja	IT22142214	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 31/08/2025

Signature of the co-supervisor: Date: 31/08/2025

Dedication

This work is dedicated to my family, friends, and mentors, whose constant support and encouragement inspired me to pursue this research. I also dedicate it to all university freshers who face the challenge of adapting to a new environment, with the hope that this project contributes in some way to making their transition smoother and more inclusive.

Acknowledgements

I would like to express my sincere gratitude to our supervisor, Ms. Narmada Gamage, and our co-supervisor, Ms. Chathurangika Kahandawarachchi, for their invaluable guidance, feedback, and encouragement throughout this project. I am also thankful to the Department of Computer Systems and Network Engineering at SLIIT for providing us with the resources and platform to carry out this research. Special thanks go to our peers and fellow students for their insightful suggestions and constructive feedback during discussions. Finally, I extend my appreciation to our families for their patience, understanding and unwavering support during this journey.

Abstract

The transition into university life often presents significant academic, social, and emotional challenges for first-year students, leading to issues of isolation, stress, and even dropout. Existing support systems tend to be static, focusing on one-time interventions or narrow academic indicators, failing to capture the dynamic needs of students. This proposal introduces UniPulse, an AI-based Social Integration and Inclusion Platform designed specifically for university freshers. Its core feature, Adaptive Student Profiling and Learning Persona Management, generates and updates multidimensional learning personas using academic, behavioral, emotional, and social interaction data. Unlike traditional static systems, UniPulse evolves with the student by integrating weekly check-ins, enabling continuous personalization and timely interventions. The research aims to design, implement, and validate the system within the academic year 2025–2026, assessing its effectiveness in improving engagement, wellbeing, and retention. By combining AI-driven analytics with ethical safeguards, UniPulse has the potential to redefine how universities support students, ensuring inclusivity, reducing attrition, and promoting overall student success.

Keywords:
Artificial Intelligence, Adaptive Profiling, Learning Personas, Student Engagement, Social Integration, Wellbeing

Table of Contents

Declaration of the Candidate & Supervisor i

Dedication ii

Acknowledgements ii

Abstract iii

Table of Contents iv

List of Figures vi

List of Tables vi

List of Abbreviations vii

List of Appendices viii

1. INTRODUCTION 1

1.1 Background 1

1.1.1 Overview of the field 1

1.1.2 Rules and regulations implemented by the government 3

1.2 Literature Survey 5

1.2.1 Research papers 5

1.3 Research Gap 8

1.4 Research Problem 10

2. OBJECTIVES 11

2.1 Main Objective 11

2.2 Specific Objectives 11

3. METHODOLOGY 13

3.1 Software Solution 13

3.2 System Architecture 16

3.3 Data Availability 19

3.3.1 Ethical clearance 19

4. PROJECT REQUIREMENTS 20

4.1 Data Collection 20

4.1.1 Functional requirements 21
4.1.2 Non-functional requirements 22
4.2 System Requirements 23
4.2.1 Software requirements 23
4.2.2 Hardware requirements 23
4.2.3 Database management 23
4.2.4 AI Model integration 23
4.2.5 Version Controlling 23
4.3 Work Break Down Structure 24
4.4 Gantt Chart 25
5. BUDGET JUSTIFICATION 26
6. COMMERCIALIZATION ASPECT 28
7. CONCLUSION 29
Reference List 30
Appendices 32
Appendix A: Input Form 1 32
Appendix B: Input Form 2 34
Appendix C: Project Timeline with Milestones 35

List of Figures

Figure 1.1: Challenges faced by university freshers [1], [2], [3], [4] 2
Figure 3.1: Overall System Architecture Diagram 16
Figure 3.2: System Architecture Diagram 18
Figure 4.1: Work Breakdown Structure 24
Figure 4.2: Gantt Chart 25

List of Tables

Table 1.1: Research Gaps 9
Table 5.1: Budget Estimation 26

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
LMS	Learning Management System
UGC	University Grants Commission
GDPR	General Data Protection Regulation
WBS	Work Breakdown Structure
WHO	World Health Organization
SaaS	Software as a Service
ASPLP	Adaptive Student Profiling and Learning Persona Management
FR	Functional Requirements
NFR	Non-Functional Requirements

List of Appendices

Appendix A: Input Form 1 32
Appendix B: Input Form 2 34
Appendix C: Project Timeline with Milestones 35

1. INTRODUCTION

1.1 Background

1.1.1 Overview of the field

The integration of artificial intelligence in student support systems has gained significant attention as a method to address the critical mental health and social integration challenges prevalent among university freshers. Research shows alarming rates of these issues, with 38% of students experiencing anxiety, 34% reporting depression symptoms [1], [2], and nearly 65% affected by social isolation [3]. These factors contribute directly to student retention, with first-year attrition rates averaging around 67% [4]. Traditional support systems struggle to meet this growing demand, leading educational institutions to seek technological solutions. Current AI-based systems focus on automated conversational interfaces and behavioral analytics, but most function as standalone tools rather than integrated platforms, limiting their effectiveness. The field continues to evolve with advancements in emotional AI and predictive analytics, creating opportunities for more sophisticated, comprehensive intervention systems.

In this report, we mainly focus on Adaptive Student Profiling and Learning Persona Management, which is the part of UniPulse that turns information such as student behavior, interactions, motivation, and emotions into continuously updated learner profiles. Unlike traditional systems that only collect data once during admission or orientation, adaptive profiling follows how a student's needs and experiences change over time. This allows the university to give timely support, whether that means academic help, wellbeing guidance, or connecting them with suitable peers.

Large-scale campus surveys consistently reveal high and fluctuating rates of mental health concerns, with significant proportions of students reporting anxiety and depression throughout their academic journey [5]. These conditions directly influence engagement and help-seeking behaviors, highlighting the necessity for dynamic, not

static, student profiles. Furthermore, loneliness is widespread, with studies indicating that approximately 64% of students experience loneliness, which often leads to disengagement and isolation [6]. Nearly 46% of educators report poorer student involvement than they did before the 2019 pandemic, indicating that engagement has also drastically decreased in the post-pandemic situation [1]. Because of these reasons, fixed profiling systems are not enough.

At the same time, evidence shows that early alert systems in higher education can positively impact student performance by improving grades and reducing withdrawal rates, strengthening the case for continuous monitoring rather than one-off interventions [7]. Additionally, research shows that personalized and adaptive learning methods increase student engagement and learning outcomes, proving the value of tailoring support to each individual [8]. Together, these results suggest that the UniPulse platform's adaptive profiling layer is essential because it converts unprocessed behavioral and emotional data into actionable, real-time insights that promote timely wellbeing outreach, tailored academic support, and targeted peer matching, all of which have a direct impact on increased student success and retention.

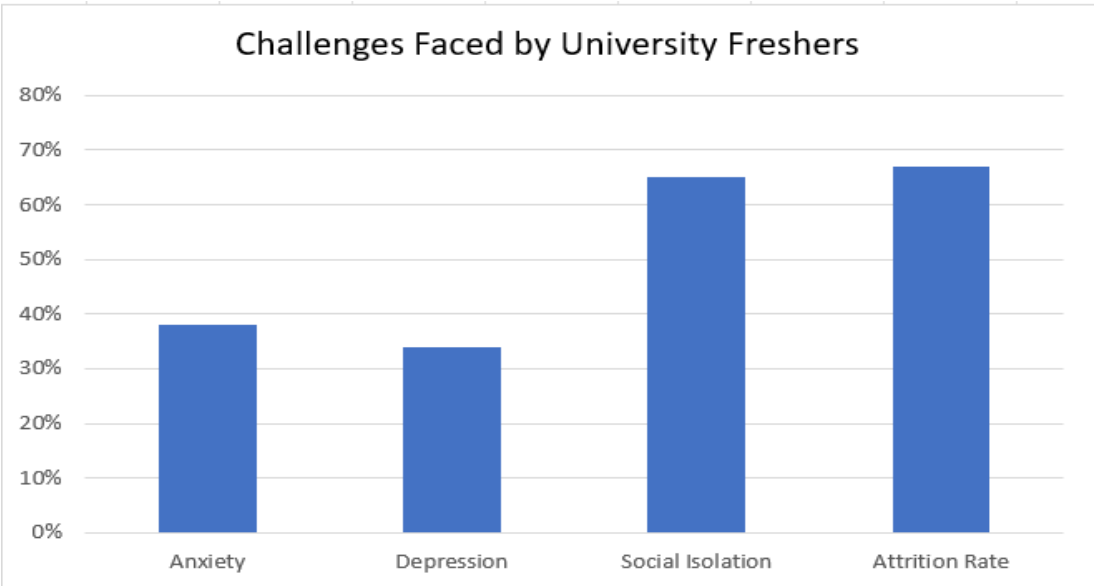


Figure 1.1: Challenges faced by university freshers [1], [2], [3], [4]

1.1.2 Rules and regulations implemented by the government

Over the years, the Sri Lankan government has introduced various policies and regulations aimed at supporting student adjustment, engagement, and wellbeing in higher education. While these initiatives were not necessarily driven by technology, they highlight the growing awareness of the challenges students face and provide important context for the need for more modern and adaptive approaches like student profiling.

One significant initiative was the Compulsory Leadership and Soft-Skills Training Program introduced in 2011 by the Ministry of Higher Education. This mandatory, residential program was designed to prepare the university entrants socially and emotionally before the start of their academic journey. Its objectives included improving employability, discipline, and integration while reducing ragging and student isolation. However, despite its initial goals, the program received criticism for its militarized approach and lack of inclusivity, leading to its eventual discontinuation around 2015. This reflects the limitations of one-time, rigid interventions that do not offer ongoing or personalized support.

The UGC has also taken steps to ensure student safety through strict anti-ragging regulations, supported by the Prohibition of Ragging and Other Forms of Violence in Educational Institutions Act, No. 20 of 1998. The need of establishing a safe atmosphere for new students is further emphasized in UGC circulars. Although these steps are necessary, they are typically legalistic and reactive rather than providing proactive or preventive ways to regularly evaluate and assist students' wellness.

In terms of broader national efforts, the National Youth Policy (2014) and the National Mental Health Policy (2005, updated in 2020) both emphasize youth development and mental wellbeing. These policies acknowledge the emotional and social challenges young people face, and they advocate for youth-friendly services and early identification of stress and depression. However, the support models still rely heavily on traditional

counseling methods, without integrating more dynamic, data-driven approaches such as continuous profiling or personalized interventions.

Sri Lanka has put in place a number of mental health programs at the school level that are in line with WHO guidelines. These programs concentrate on helping teenagers develop their coping mechanisms, emotional literacy, and resilience. However, surveys show that a significant number of students experience loneliness in spite of these initiatives, suggesting that many emotional problems go untreated. In addition, there is a lack of continuity in mental health assistance during the school-to-university transition, and there aren't many mechanisms in place to track or adjust to students' changing needs when they start college.

These existing efforts show a strong foundation and a clear recognition of the issues at hand. However, they also highlight a gap in sustainable, personalized, and technology-driven support systems, underscoring the relevance and timeliness of introducing adaptive student profiling and persona-based interventions through platforms like UniPulse.

1.2 Literature Survey

1.2.1 Research papers

Transitioning into university life can be emotionally exhausting for many first-year students, especially those coming from unfamiliar academic and social environments. Traditional support systems often lack the responsiveness and personalization needed to address the challenges faced by today's learners. Numerous studies have been conducted on this subject.

1. **"Personalized adaptive learning in higher education: A scoping review of key characteristics and impact on academic performance and engagement" (2024) by E. du Plooy, Daleen Casteleijn and Denise Franzsen -**
   This review examined adaptive learning implementations across universities, with a focus on learner modeling, personalized feedback, and real-time adjustment. The findings showed that around 36% of studies reported improved student engagement via adaptive feedback, and 59% noted enhanced academic outcomes. These results highlight the growing significance of personalized learning systems in shaping university-level education. [8]

2. **"The Impact of Artificial Intelligence on Personalized Learning within Higher Education: A Systematic Review" (2025) by Carlos Merino-Campos-**
   In accordance with PRISMA principles, this systematic review examined AI applications in higher education, including knowledge tracing, sentiment analysis, and recommender systems. The study demonstrated the possibility of real-time adaptive procedures in academic contexts and confirmed that AI-based personalization plays an essential role in improving learner engagement and performance. [9]

3. **"Artificial Intelligence-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education" (2023) by Ramteja Sajja, Yusuf Sermet, Muhammed Cikmaz, David Cwiertny and Ibrahim Demir –**
   This paper introduced the AIIA system, which integrates natural language processing and AI to deliver personalized tutoring, flashcards, and quizzes within learning management systems. The study illustrated how these intelligent assistants may offer quick adaptive assistance, reducing learning challenges and improving engagement. [10]

4. **"Mitigating Biases in Student Performance Prediction via Attention-Based Personalized Federated Learning" (2022) by Yun-Wei Chu, Seyyedali Hosseinalipour, Elizabeth Tenorio, Laura Cruz, Kerrie Douglas, Andrew Lan, Christopher Brinton –**
   The goal of this study was to develop more accurate and fair learner models by utilizing federated learning and attention mechanisms. In order to demonstrate how personalized integrated learning can enhance performance prediction while preserving data privacy across several student groups, the study analyzed multimodal behavioral data. [11]

5. **"AI-enabled Personalized Learning: Empowering Management Students for Improving Engagement and Academic Performance" (2024) by Adil Ellikkal & S. Rajamohan-**
   This study examined how AI-driven personalization improved management students' autonomy, competence, and relatedness. It was based on the Self-Determination Theory. The findings clearly improved academic performance and engagement, highlighting the function of adaptive systems in encouraging student motivation and success in higher education. [12]

6. **“The relationship between faculty interactions, sense of belonging, and academic stress: a comparative study of the post-COVID-19 college life of Korean and international graduate students in South Korea” by Dongil Kim, Yeyoung Woo, Jusuk Song, Subin Son (2023)-**

   This study, carried out among graduate students (both Korean and international), examined how interactions with faculty affect academic stress through a sense of belonging. The findings showed that stronger faculty interactions lead to a greater sense of belonging, which in turn significantly reduce academic stress. The study also found that international students experience a greater effect from faculty interactions on stress. [13]

1.3 Research Gap

1. **Reliance on Static or Narrow Data Sources [8], [10]-**
   Standardized tests, attendance, and grades are examples of static learner records that are used by many current systems. Although these offer certain academic perspectives, they fall short in capturing the social, emotional, and motivational fluctuations that have a big influence on first-year students. Our approach fills this gap by adding multimodal data to the profiling process, including sentiment, interaction behavior, and social engagement patterns.

2. **Limited Integration of Emotional and Motivational Indicators [8], [11]-**
   The majority of adaptive learning implementations continue to concentrate on academic personalization and cognitive outcomes, despite new research showing its advantages. Learner personas, which are crucial for promoting the overall wellness of students, rarely incorporate emotional or motivational data. By creating learning personas that adjust to both academic and emotional shifts, UniPulse directly addresses this issue.

3. **Short-Term, Small-Scale Pilots Without Scalability [12], [13]-**
   Most of the research has been done with small trial groups or early-stage programs. These solutions usually don't have the infrastructure needed to scale across whole institutions or a variety of student populations. Our solution can be widely implemented for incoming university freshmen because it was created with scalability and real-time adaptation in mind.

4. **Absence of Preventive Measures [10], [11]-**
   Existing early-alert systems often identify academic risk but rely heavily on human follow-up (advisors, instructors). Interventions may come too late because of this reactive strategy. UniPulse, on the other hand, employs adaptive

profiling to provide real-time wellbeing reminders, peer connections, and personalized nudges in addition to identifying hazards early.

5. **Ethical Considerations Understudied [9], [11]-**

   Current adaptive profiling methods often disregard privacy, openness, and bias. There is a chance that this will increase gaps in education. In order to guarantee equity and student trust, UniPulse specifically incorporates privacy-preserving measures and bias reduction techniques.

Table 1.1: Research Gaps

Features	[8]	[9]	[10]	[11]	[12]	[13]	UniPulse
Continuous updates (Adaptive profiling)	✗	✗	✓	✗	✗	✗	✓
Behavioral usage analytics	✗	✗	✓	✓	✓	✗	✓
Educational & motivational mapping	✗	✗	✗	✗	✓	✗	✓
Personalized feedback	✓	✗	✗	✗	✗	✗	✓
Social interaction analytics	✗	✗	✗	✗	✗	✓	✓
Cognitive, emotional integrated profiles	✗	✓	✗	✗	✗	✗	✓

In summary, while existing studies demonstrate progress in adaptive learning and AI personalization, they remain limited to academic or behavioral data, lack consistent persona updates, and overlook the integration of emotional, motivational, and social factors. UniPulse bridges these gaps by creating a unique, evolving student persona that combines cognitive and emotional dimensions, updated weekly, and supported by decision modeling and social analytics. This ensures that first-year students receive more holistic, personalized, and timely support than has been possible in prior research.

1.4 Research Problem

- **What is the adaptive profiling mechanism for dynamically modeling university students?-** The static student profiles that are currently in place were made at orientation with limited academic and demographic information. They fall short of capturing a student's changing social, emotional, and academic path. In order to create and update comprehensive student profiles, our solution presents a dynamic, AI-driven profiling system that analyzes multi-modal data, including weekly check-ins, platform interactions, and academic activities. This makes it possible to move from a static image to a dynamic movie of every student's experience at university.

- **How can multi-dimensional student data be effectively integrated into the profiling system?-** An initial onboarding form and simple weekly check-ins intended for high compliance are used to collect student data. This organized data includes motivational factors, emotional state, social comfort, and academic confidence. This self-reported data is combined with behavioral analytics by the system. All data is handled strictly in accordance with ethical data protection standards and anonymized for analysis.

- **How will the success of the adaptive student profiling system be evaluated?-** Success will be measured through a combination of technical accuracy and impact metrics. This covers the association between persona alterations and actual academic/social outcomes, weekly check-in completion rates (>80%), and persona accuracy (as determined by student self-validation >85%). Surveys conducted before and after the intervention will evaluate benefits in self-reported stress reduction, confidence, and belonging. A staged pilot study will be used to validate the system, and iterative improvements will be made in response to advisor and student input.

2. OBJECTIVES

2.1 Main Objective

To design and implement an AI-driven Adaptive Student Profiling and Learning Persona Management system that generates and updates multidimensional student personas on a weekly basis, by integrating behavioral, emotional, motivational, and social interaction data, in order to provide personalized and timely support for university freshers within the academic year 2025–2026.

2.2 Specific Objectives

1. **Behavioral and Academic Data Analysis-**
   To gather and examine information about how students use the platform, including learning activities, frequency of access, and navigations patterns. This will support the development of the behavioral and cognitive layer of adaptive student persona.

2. **Emotional and Motivational Profiling-**
   To use sentiment analysis and engagement scoring to combine motivational and emotional cues. This will enable student personas to represent not only academic achievement but also motivation and well-being.

3. **Decision Modeling for Personalization-**
   To design and implement a decision-modeling framework that adjusts student personas and initiates suitable interventions, such as wellbeing support, peer recommendations, and encouragement.

4. **Social Interaction Analytics-**

   To incorporate analytics on peer interactions and group activities. This will help capture social engagement and belonging as part of the learning persona.

5. **Weekly Adaptive Persona Updates-**

   To ensure that learner personas evolve on a weekly basis, reflecting current trends in student behavior, motivation, and emotional state.

6. **System Validation and Evaluation-**

   To test the effectiveness of the proposed system in identifying at-risk students and providing timely, personalized support, and to compare its outcomes with existing static profiling approaches.

3. METHODOLOGY

3.1 Software Solution

The Adaptive Student Profiling and Learning Persona Management module stands as a core feature of the UniPulse platform, developed to collect, analyze, and continuously update individualized student personas. Structured yet highly adaptive, this module provides a dynamic understanding of each student by integrating self-reported data into personalized learning and wellbeing profiles. It primarily serves the students.

The process begins when a university fresher registers on the UniPulse platform. Data are collected from the students through two forms one which is given when onboarding and the other given weekly:

1. **Initial Student Profiling Form** (Appendix A)

This comprehensive form captures a wide range of data spanning four key domains:

- Academic & Demographic Details

  Includes the student's name, university and faculty, A/L stream and results, preferred learning style, and their self-rated confidence in transitioning to university life.
- Social & Personality Traits

  Explores social preferences, introversion/extroversion tendencies, stress levels, comfort with group work, and preferred communication modes.
- Interests & Lifestyle

  Covers motivators, clarity of personal and academic goals, hobbies, current living arrangements, and employment status.
- Wellbeing & Support Needs

  Assesses potential emotional challenges, perceived barriers to inclusion, preferred support methods, and invites open-ended reflections.

Once submitted, the data is analyzed by the platform's AI engine to generate an initial learning persona. This persona represents a dynamic baseline model that synthesizes academic background, social behaviors, emotional readiness, and motivational drivers.

Rather than being a static profile, it serves as a starting point for continuous updates throughout the student's university journey.

Recognizing that students' needs and experiences evolve, particularly during major life transitions like entering university the system integrates a Weekly Student Check-In Form to ensure regular and meaningful updates.

2. **Weekly Student Check-In Form** (Appendix B)

This brief yet insightful questionnaire is divided into two sections:

- Emotional & Mental Wellbeing

  Students rate their current mood, sleep quality, focus, peer support, openness to receiving help, and reflect on recent experiences.
- Social & Academic Behavior

  Captures levels of peer interaction, participation in group activities, current academic workload, and any notable challenges.

These weekly submissions allow the AI-driven analytics engine to track real-time emotional, social, and academic trends. The data feeds into the existing learning persona, ensuring it remains an up-to-date and accurate representation of the student's current state. For instance, a gradual increase in reported confidence and peer engagement may signal positive adaptation, whereas persistent stress or social withdrawal may trigger early intervention protocols.

Once data is collected, UniPulse's analytics engine processes and interprets key behavioral, emotional, and cognitive signals. This facilitates the creation of adaptive learning personas that evolve in response to each student's journey. The platform's decision modeling layer ensures that profile changes translate into meaningful actions:

- Students flagged for low social engagement may be prompted with targeted peer-matching suggestions.
- Those showing signs of stress or academic fatigue may be directed to relevant wellness resources or referred to support staff.

- High-performing, socially active students may be highlighted for leadership or mentorship opportunities.

All collected data is securely stored in the UniPulse database in line with strict privacy and ethical standards. Students are granted access to simplified visualizations of their evolving persona, giving a sense of self-awareness, and reflection in their academic and social development.

The Adaptive Student Profiling and Learning Persona Management module transforms what would otherwise be static data into a responsive system that supports continuous understanding of each student. By integrating academic, emotional, and social dimensions into one adaptive framework, UniPulse ensures that support is not just timely but also personalized.

3.2 System Architecture

The overall system architecture diagram of UniPulse illustrates how the four major components, Adaptive Profiling, Peer Matching, Risk Detection, and Conversational Support interact to create a holistic student support ecosystem. The student acts as the primary user, providing academic, emotional, and social inputs which are stored in a secure database. These inputs are processed by AI-driven analysis, which then powers the four interconnected modules. Each module contributes to improving student wellbeing and integration: Adaptive Profiling builds evolving personas, Peer Matching connects compatible students, Risk Detection identifies early signs of social or emotional difficulties, and Conversational Support provides real-time wellbeing assistance. Together, the architecture highlights the system's role as a comprehensive integration and inclusion platform.

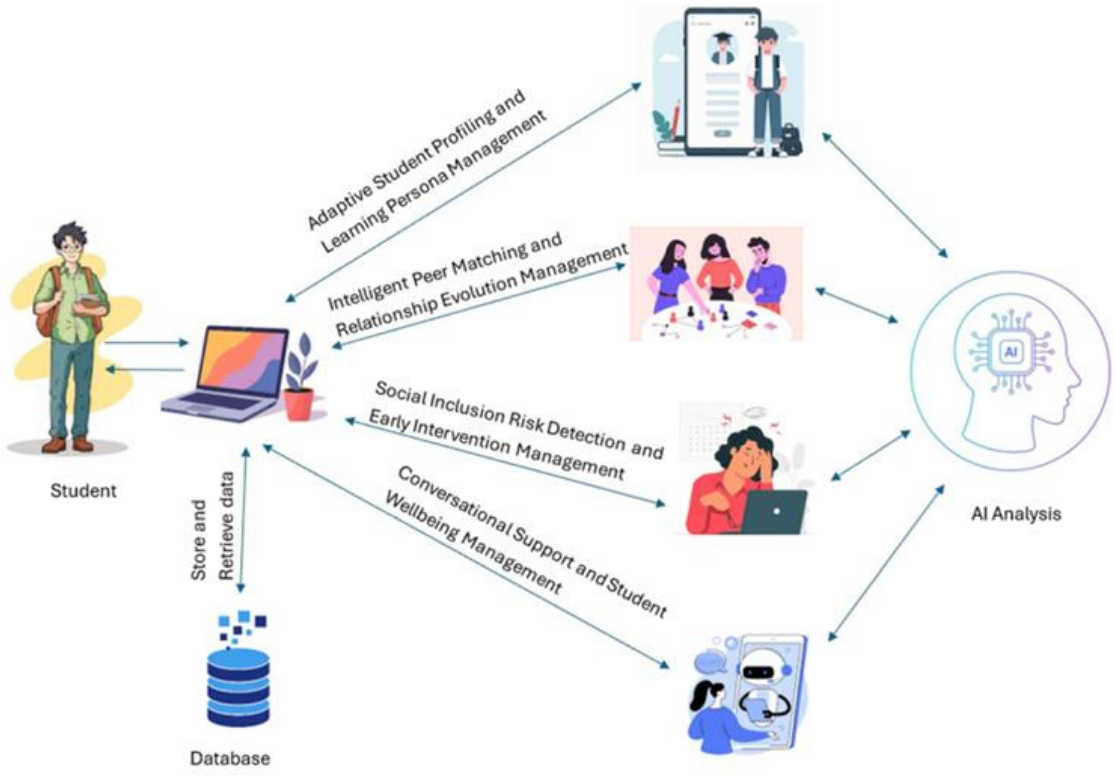


Figure 3.1: Overall System Architecture Diagram

The system architecture for the Adaptive Student Profiling and Learning Persona Management component, depicted in Figure 3.2, is engineered to create and maintain dynamic, multidimensional student profiles. Data collection is the first step in the process, when students supply academic, demographic, and wellness data during the first onboarding stage. Weekly check-in forms, which record varying emotional states, motivational levels, and social interaction patterns, continuously enhance this core data. In order to ensure consistency and dependability, this raw, multi-modal data stream goes through a preprocessing step that cleans, normalizes, and structures it for analysis.

The preprocessed data is then consumed by the AI Profiling Engine, the core analytical layer of the architecture.

The preprocessed data is then fed into the AI Profiling Engine, which serves as the core analytical component of the architecture. This engine builds a thorough understanding of every student by combining several analytical techniques. Behavioral Analytics examines how students interact with the platform and engage in academic tasks. Emotional and Motivational Analysis applies sentiment analysis to qualitative responses to evaluate wellbeing and motivation levels. Academic and Social Tracking connects social connectedness, academic performance, and engagement data. These insights are combined by the engine's Persona Generator, which creates a comprehensive and flexible learning persona using clustering and classification models. This persona serves as a live model that is updated weekly and is always changing to reflect the student's emotional and academic development.

Finally, these dynamically updated personas are persisted in a centralized Student Profile Database, which acts as the single source of truth for student identity within the UniPulse platform. Through secure APIs, the personas are made available to other components, including conversational support interfaces, peer-matching algorithms, and early risk detection. This makes it possible for all other platform components to provide highly customized and context-aware interventions and support. This architecture immediately tackles the significant drawback of conventional, static profiling systems by

guaranteeing that student profiles are comprehensive and constantly improved, enabling timely and relevant support throughout the student's university journey.

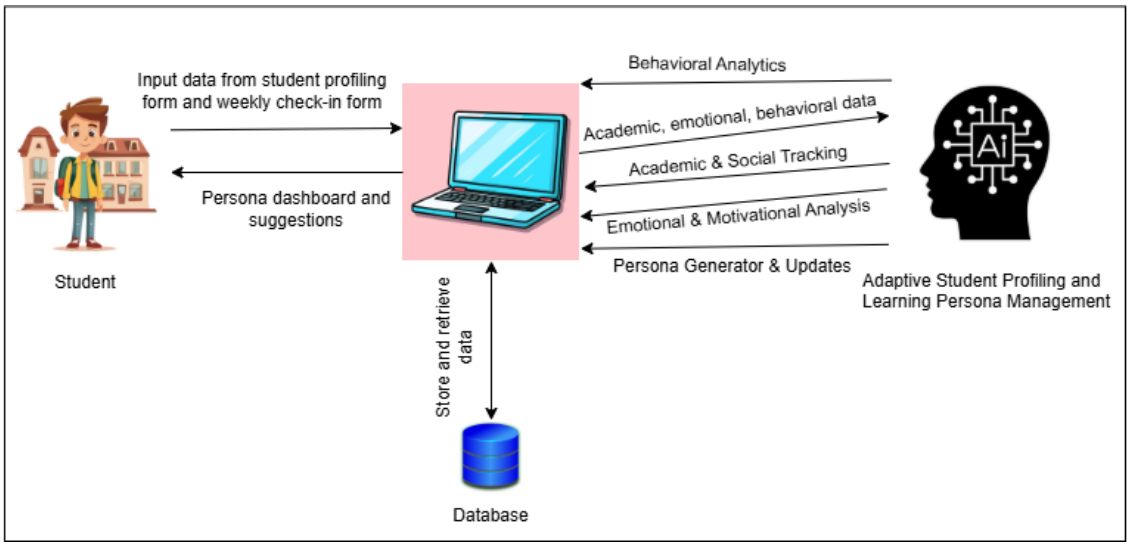


Figure 3.2: System Architecture Diagram

3.3 Data Availability

3.3.1 Ethical clearance

Ethical clearance for this research project, titled "AI-Based Social Integration and Inclusion Platform for University Freshers," has been secured through the SLIIT Ethics Review Committee's expedited procedure. The study involves collecting sensitive, individually identifiable data from first year, first semester students through comprehensive surveys, onboarding profiles, and weekly emotional check-ins. All data collection will proceed only after obtaining explicit informed consent, with transparent communication about usage protocols and privacy safeguards. Individually identifiable data will be encrypted and stored on secure institutional servers with rigorous access controls. Upon project completion, all raw data containing personal identifiers will be permanently deleted from all systems and devices to ensure participant privacy and confidentiality.

Specifically, regarding the Adaptive Student Profiling and Learning Persona Management subsystem, this component processes comprehensive data gathered from first-year students through structured onboarding forms and weekly check-ins. The collected information includes academic and demographic details, social and personality traits, lifestyle interests, and wellbeing indicators. Weekly updates capture fluctuations in emotional state, academic focus, peer interactions, and support needs, allowing the system to dynamically refine student learning personas over time. All collected data is managed using coded identifiers and secured under the same ethical and privacy protocols governing the broader research project. No raw personal identifiers will be exposed within the profiling outputs. Upon completion of the project, all identifiable datasets, including onboarding and check-in responses, will be permanently deleted to ensure full confidentiality and adherence to institutional ethical standards.

4. PROJECT REQUIREMENTS

4.1 Data Collection

At the initial stage, data will be collected through two structured Google Forms distributed to participating first-year university students:

- **Initial Profiling Form:** Completed once during onboarding, capturing demographic, academic, social, lifestyle, and wellbeing details. This provides a baseline profile for generating the first learner persona.
- **Weekly Check-In Form:** Captures emotional wellbeing, social interactions, academic focus, and engagement signals on a weekly basis. This includes validated measures such as DASS-21, PHQ-9, MBI-SS, GHQ-12, and UCLA Loneliness subscales to ensure evidence-based tracking of student wellbeing and inclusion.

In later phases of system implementation, these forms will be integrated directly into the UniPulse application instead of relying on external Google Forms. Students will complete their weekly check-ins as part of the login process at the beginning of each week, ensuring consistent and timely updates to their profiles. All collected data will be securely stored, preprocessed, and fed into the profiling algorithms for persona generation and weekly updates.

4.1.1 Functional requirements

FR-ASPLP-1: Completion of Initial Profiling Form

At onboarding, students complete the Initial Profiling Form once to establish their baseline persona.

FR-ASPLP-2: System Login & Weekly Check-In Prompt

When students log in for the first time in a given week, the system must prompt them to complete the Weekly Check-In form before accessing other features.

FR-ASPLP-3: Weekly Persona Updates

The system must update student personas weekly based on check-in responses, ensuring profiles reflect changing needs.

FR- ASPLP-4: AI-Based Persona Generation

AI models process both initial and weekly data (academic, emotional, behavioral, and social) to generate and refine multidimensional student personas.

FR- ASPLP-5: Personalized Feedback & Recommendations

The system offers suggestions including academic reminders, wellbeing prompts, and peer-matching recommendations following every update.

FR- ASPLP-6: Data Transition from Google Forms to In-App Forms

The system must support phased integration, starting with external Google Forms and later migrating into built-in UniPulse forms for direct data capture.

4.1.2 Non-functional requirements

NFR-ASPLP-1: Availability

The platform should maintain high availability with minimal downtime, allowing students to complete check-ins consistently.

NFR-ASPLP-2: Performance

The system should provide fast response times when loading dashboards, forms, and recommendations.

NFR-ASPLP-3: Reliability

Data collection, persona updates, and recommendations must be accurate and consistent across all users.

NFR-ASPLP-4: Scalability

The system should be able to handle large numbers of concurrent users without performance degradation.

NFR-ASPLP-5: Usability

The platform interface should be intuitive and user-friendly, ensuring easy navigation for students with different levels of technical skills.

NFR-ASPLP-6: Security & Privacy

Sensitive student data should be encrypted, access-controlled, and protected by GDPR-like standards to ensure ethical handling of personal information.

4.2 System Requirements

4.2.1 Software requirements

- Laravel 11 (Blade + Bootstrap) – For core application development.
- MySQL – For structured storage of student, profile, and check-in data.
- Git/GitHub – For version control and collaborative development.

4.2.2 Hardware requirements

- Minimum Intel i5/Ryzen 5 processor
- 16GB RAM
- 500GB SSD
- Linux/Windows OS

4.2.3 Database management

UniPulse will use MySQL as the database system. Key aspects include:

- Data Security: Encryption and role-based access control.
- Reliability: Daily backups and disaster recovery mechanisms.
- Scalability: Indexed queries and optimized schemas to support large student populations.
- Compliance: GDPR-like standards to ensure ethical handling of sensitive student data.

4.2.4 AI Model integration

The AI component is central to adaptive profiling. It integrates:

- Sentiment Analysis: Detecting emotions through weekly inputs.
- Engagement Analytics: Measuring participation and academic focus.
- Decision Modeling: Recommending personalized support actions.

4.2.5 Version Controlling

Version control will be handled using Git and GitHub, following the GitFlow branching strategy. This allows structured collaboration, pull requests, and code reviews.

4.3 Work Break Down Structure

The Figure 4.1 represents the Work Breakdown Structure (WBS) for the “UniPulse” Web Application project, breaking down the entire scope of work into manageable phases and tasks. It is structured into seven main phases: Background Study, Requirements, Documentation, Design, Implementation, Testing, and Deployment. Each phase is further subdivided into specific deliverables or activities, such as feasibility study, requirements analysis, UI design, AI processing, and system testing. This hierarchical breakdown helps in organizing the project workflow, assigning responsibilities, estimating timelines, and ensuring that all aspects of the project are thoroughly planned and executed in a structured manner.

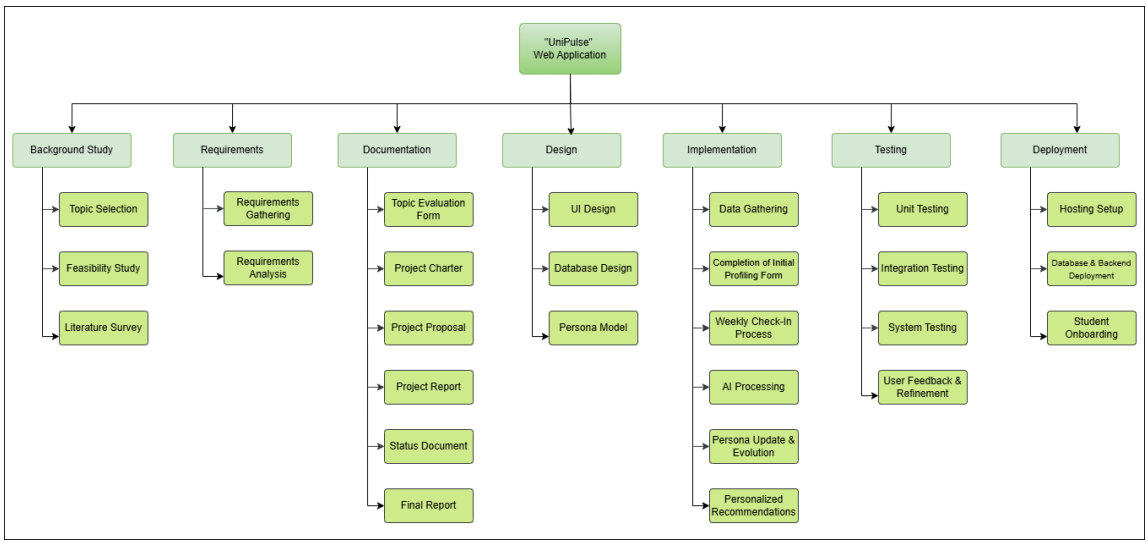


Figure 4.1: Work Breakdown Structure

4.4 Gantt Chart

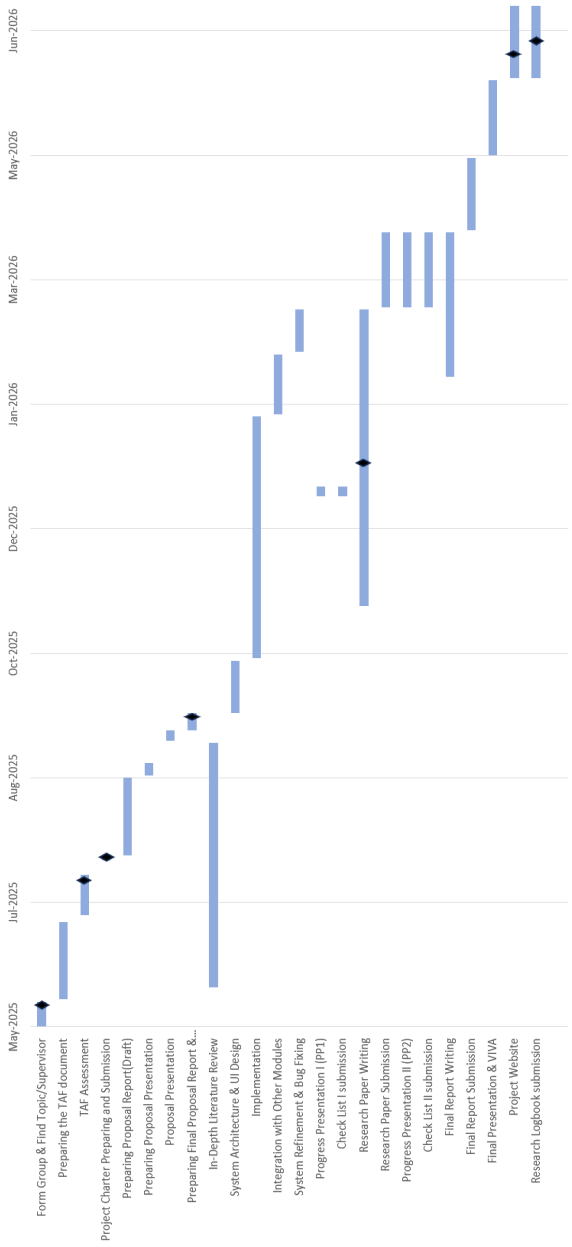


Figure 4.2: Gantt Chart

The Figure 4.2 presents a structured timeline for the Adaptive Student Profiling and Learning Persona component from May 2025 to June 2026. It begins with group formation, topic selection, and proposal development, followed by an extended literature review and system design phase.

Core development takes place from October to February, including implementation, module integration, and system refinement. Progress presentations and checklist submissions are placed to track key milestones.

All key contributions, including the final report, presentation, website, and logbook, are scheduled between April and June 2026. The last phase is centered on documentation, and report writing.

5. BUDGET JUSTIFICATION

The UniPulse AI-Based Social Integration and Inclusion Platform is expected to cost Rs. 725,000.00 in total. To guarantee strong development, seamless acceptance, and excellent usability, the budget is distributed among certain operational, technical, and human resource components. System development, interface and design, personnel expenses, testing, and support activities are all included in the breakdown.

Table 5.1: Budget Estimation

Cost Area	**Description**	**Estimated Cost (Rs.)**
Backend Development	Building the platform's main system, AI features, and database.	180,000.00
Frontend Development	Creation of responsive and interactive user interfaces enhancing usability and accessibility for university freshers.	80,000.00
UI/UX Design & Prototyping	Design layouts, wireframes, prototyping, user testing.	90,000.00
Developers & Technical Support Staff	Salaries for developers, AI engineers, support staff.	150,000.00
Project Management	Planning, coordination, and documentation.	80,000.00
Training & Awareness Workshops	Conducting awareness workshops and providing user guides for students and staff.	30,000.00
AI Model and	Developing AI models and algorithms	25,000.00

Algorithm		
Testing & Quality Assurance	Functional testing, bug fixing, performance validation	40,000.00
Maintenance & Minor Updates	Post-deployment fixes, small improvements	50,000.00
Total Estimated Cost		**725,000.00**

6. COMMERCIALIZATION ASPECT

The commercialization of UniPulse is centered on positioning it as a scalable, cost-effective wellbeing and integration platform for universities. The primary market consists of institutions facing high dropout rates and growing demand for mental health support, particularly for first-year and international students, introverts, and those from rural or unfamiliar environments who may fear organizations or struggle to adjust. These groups are most vulnerable to isolation, stress, and disengagement, making them the core focus of UniPulse. Extroverted students may also benefit for self-reflection and support, though they are not the primary target. UniPulse will be marketed through direct institutional partnerships, pilot projects, and presence at educational technology forums, highlighting its ability to reduce attrition and enhance student satisfaction.

Implementation follows a phased approach, beginning with a faculty-level pilot to demonstrate impact and gather feedback, followed by institution-wide rollout and eventual integration with the university's LMS and student portals. This staged deployment ensures smooth adoption while allowing the platform to adapt to institutional needs. The financial model is subscription-based (SaaS), where universities pay either a yearly license fee or a per-student subscription. Tiered pricing options will allow institutions to choose between core wellbeing monitoring and premium analytics with counselor integration.

From a financial perspective, UniPulse offers clear returns by reducing student attrition, which preserves tuition revenue and enhances institutional reputation. Even a small reduction in dropout rates translates into significant savings and improved rankings. Operating costs mainly cover AI system maintenance, cloud hosting, data security, and customer support, making the model sustainable at scale. UniPulse's competitive edge lies in its proactive, AI-driven nature: unlike traditional counseling services that react after problems occur, the platform continuously monitors trends, provides personalized support, and escalates serious cases automatically extending wellbeing support to every student without proportional increases in staffing costs.

7. CONCLUSION

A prompt and creative solution to the increasing difficulties experienced by university freshers in adjusting to new academic and social contexts is suggested by the UniPulse-AI-Based Social Integration and Inclusion Platform. The diverse and evolving needs of students can no longer be met by traditional support systems alone, since anxiety, sadness, and social isolation are increasing. Behavioral analytics, intelligent peer matching, adaptive student profiles, and conversational support are all integrated by UniPulse to provide a comprehensive and data-driven approach to student engagement and wellbeing.

Through the use of modern technologies like AI-based profiling and real-time analytics, the system offers individualized, dynamic support that changes as each student progresses. This guarantees not only academic achievement but also enhanced mental health, more solid social ties, and lower attrition rates. With an emphasis on accessibility, scalability, and inclusion, the project's design is flexible enough to be used in a variety of academic settings while yet being mindful of ethical and student privacy issues.

In conclusion, by providing institutions with a proactive, tech-driven tool to promote integration, inclusion, and resilience among freshers, UniPulse has the potential to transform student support services. Through this platform, universities can create a more supportive and inclusive academic community, ultimately contributing to better retention, improved student outcomes, and healthier campus environments.

Reference List

[1] E. K. Fittes, "Student Engagement in Learning Is Low. A Survey Offers Clues on How to Bring Them Back," EdWeek Market Brief, 24 October 2024. [Online]. Available: https://marketbrief.edweek.org/meeting-district-needs/student-engagement-in-learning-is-low-a-survey-offers-clues-on-how-to-bring-them-back/2024/10. [Accessed 18 August 2025].

[2] J. Publications, "Therapeutic Potential of Social Chatbots in Alleviating Loneliness and Social Anxiety: Quasi-Experimental Mixed Methods Study," Journal of Medical Internet Research, 14 January 2025. [Online]. Available: https://www.jmir.org/2025/1/e65589. [Accessed 18 August 2025].

[3] EDUCAUSE, "2024 EDUCAUSE Horizon Report | Teaching and Learning Edition," EDUCAUSE, 13 May 2024. [Online]. Available: https://library.educause.edu/resources/2024/5/2024-educause-horizon-report-teaching-and-learning-edition. [Accessed 18 August 2025].

[4] A. Eser, "Student Loneliness Statistics," ZIPDO, 30 May 2025. [Online]. Available: https://zipdo.co/student-loneliness-statistics/. [Accessed 18 August 2025].

[5] A. Mowreader, "Student Wellness Tip: Supporting Lonely Learners," 23 May 2024. [Online]. Available: https://www.insidehighered.com/news/student-success/health-wellness/2024/05/23/report-addressing-loneliness-college-campuses. [Accessed 21 August 2025].

[6] J. Alonso, "Study: Academic Alerts Slash Course Withdrawal Rates by 30%," Inside Higher Ed, 22 January 2025. [Online]. Available: https://www.insidehighered.com/news/students/retention/2025/01/22/research-shows-academic-alerts-impact-grades-withdrawals. [Accessed 21 August 2025].

[7] E. d. Plooy, D. Casteleijn and D. Franzsen, "Personalized adaptive learning in higher education: A scoping review of key characteristics and impact on academic performance and engagement," Heliyon, 15 November 2024. [Online]. Available:

https://www.sciencedirect.com/science/article/pii/S2405844024156617. [Accessed 21 August 2025].

[8] C. Merino-Campos, "The Impact of Artificial Intelligence on Personalized Learning in Higher Education: A Systematic Review," MDPI, 26 March 2025. [Online]. Available: https://doi.org/10.3390/higheredu4020017. [Accessed 21 August 2025].

[9] R. Sajja, Y. Sermet, M. Cikmaz, D. Cwiertny and I. Demir, "Artificial Intelligence-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education," arXiv, 19 September 2023. [Online]. Available: https://arxiv.org/abs/2309.10892. [Accessed 22 August 2025].

[10] S. H. Yun-Wei Chu, E. Tenorio, L. Cruz, K. Douglas, A. Lan and C. Brinton, "Mitigating Biases in Student Performance Prediction via Attention-Based Personalized Federated Learning," arVix, 2 August 2022. [Online]. Available: https://arxiv.org/abs/2208.01182. [Accessed 22 August 2025].

[11] A. Ellikkal and S. Rajamohan, "AI-enabled personalized learning: empowering management students for improving engagement and academic performance," Emerald Insight , 24 July 2024. [Online]. Available: https://www.emerald.com/xjm/article/22/1/28/1273239/AI-enabled-personalized-learning-empowering. [Accessed 22 August 2025].

[12] H. M. Publications, "THE HEALTHY MINDS STUDY 2023-2024 Data Report," 2024. [Online]. Available: https://healthymindsnetwork.org/wp-content/uploads/2025/04/2023-2024-HMS-National-Data-Report_041525.pdf. [Accessed 21 August 2025].

[13] D. Kim, Y. Woo, J. Song and S. Son, "The relationship between faculty interactions, sense of belonging, and academic stress: a comparative study of the post-COVID-19 college life of Korean and international graduate students in South Korea," Frontiers in Psychiatry, Vols. Volume 14 - 2023, 2023.

Appendices

Appendix A: Input Form 1

Student Profiling Form

* Indicates required question

Academic & Demographic Details Untitled title

Name *

Your answer

University *

- Sri Lanka Institute of Information Technology(SLIIT)
- NSBM Green University
- Informatics Institute of Technology(IIT)
- University of Colombo
- University of Sri Jayewardenepura
- University of Kelaniya
- Other:

Faculty *

- Faculty of Computing
- Faculty of Engineering
- Faculty of Business
- Faculty of Science
- Faculty of Humanities & Social Sciences
- Other:

AL Stream *

- Bio Science
- Physical Science
- Arts
- Commerce
- Other:

AL Results *

	A	B	C	S	F
Subject 1					
Subject 2					
Subject 3					
English					
General Knowledge					

What is your preferred learning style *

- Online
- Physical
- Hybrid

How confident are you in transitioning to university life? *

	1	2	3	4	5	
Not confident at all						Extremely confident

Social & Personality Traits

Which social setting do you prefer most? *

- 1-on-1 interactions
- Small Groups
- Large Groups
- Online-only
- Other:

Where would you place yourself on the Introvert–Extrovert scale? *

	1	2	3	4	5	6	7	8	9	10	
Fully Introvert											Fully Extrovert

In general, how would you describe your usual level of stress or anxiety? *

- Low
- Moderate
- High

How comfortable are you working in group activities? *

	1	2	3	4	5	
Not comfortable at all						Very comfortable

Which communication methods do you prefer? (Select all that apply) *

- Texts
- Calls
- In-person conversations

Next

Clear form

Interests & Lifestyle

What motivates you most about university life? *

- Academic growth
- Career opportunities
- Friends and connections
- Experiences and exposure
- Other:

I have a clear goal or purpose for my university journey.* *

	1	2	3	4	5	
Strongly disagree						Strongly agree

What are your top interests? *

- Sports
- Arts
- Technology
- Reading
- Social events
- Other:

What are your favorite hobbies or activities in your free time?

- Reading
- Watching Dramas
- Sports
- Painting
- Traveling
- Volunteering
- Gaming
- Listening to music
- Other:

What is your current living arrangement? *

- Hostel
- Home
- Boarding place
- Other:

Are you currently employed? *

- Yes
- No

Wellbeing & Support Needs

I often feel overwhelmed or anxious. *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

I struggle to connect with peers. *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

I would use an AI platform for wellbeing support. *

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	Strongly agree

Which types of support would you find helpful? *

- [ ] Peer matching
- [ ] Counseling services
- [ ] Study groups
- [ ] AI chatbot support

Back Submit Clear form

Appendix B: Input Form 2

Weekly Check-In Form

* Indicates required question

Emotional & Mental Wellbeing

This week, my overall mood was *

	1	2	3	4	5	
Very negative	O	O	O	O	O	Very positive

I felt supported by my peers/university this week *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

What emotion best describes your week? *

- Anxious
- Calm
- Lonely
- Excited
- Overwhelmed
- Neutral
- Other

I had trouble sleeping because of stress or thoughts *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

I found it hard to stay focused on academic tasks *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

I would be open to talking to a mentor or counselor if I needed help *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

I know how to access mental health support if I needed it *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

I've been feeling tense or unable to relax. *

	1	2	3	4	5	
Never	O	O	O	O	O	Always

I've been worrying about many things lately. *

	1	2	3	4	5	
Not at all	O	O	O	O	O	Very much

Social & Academic Behavior

How often did you interact with peers outside class this week? *

	1	2	3	4	5	
Not at all	O	O	O	O	O	Daily

"I was able to keep up with my academic workload this week" *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

Which group activities did you participate in this week? *

- [ ] Study groups
- [ ] Social events
- [ ] Clubs
- [ ] None

What academic challenges did you face this week? *

- Assignments
- Exams
- Group work
- Understanding lessons
- Time management
- Other
- None

I often feel left out or disconnected from others. *

	1	2	3	4	5	
Never	O	O	O	O	O	Always

I feel like I don't have anyone to talk to when I'm struggling *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

Depressive Feelings & Low Mood

I often feel like I have no energy or motivation. *

	1	2	3	4	5	
Never	O	O	O	O	O	All the time

I find little pleasure or enjoyment in things I used to like. *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

I've been feeling down, hopeless, or sad most of the time. *

	1	2	3	4	5	
Not at all	O	O	O	O	O	Extremely

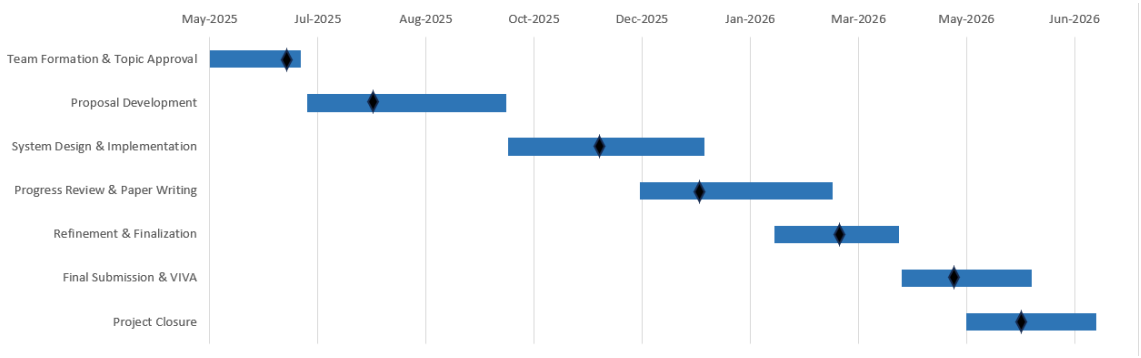
Disengagement & Burnout

I feel emotionally drained by my studies. *

	1	2	3	4	5	
Not at all	O	O	O	O	O	Completely

I feel like I'm just going through the motions without interest. *

	1	2	3	4	5	
Strongly disagree	O	O	O	O	O	Strongly agree

Submit Clear form

Appendix C: Project Timeline with Milestones